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Electromagnetic Fault Injection : how faults occur ?

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***Abstract*—Electromagnetic Fault Injection (EMFI) has recently gained popularity as a mean to induce faults because of its inherent advantages. Among them, the most interesting is probably its ability to generate faults in Systems on Chips without removing the package, and this even if only the frontside is exposed to the EM field.**

Despite this popularity, there is only little information on how EMFI generates faults. Within this context, this paper first aims at filling this lack by proposing a complete modeling of EM induction fault mechanism. In a second step, the introduced model is confronted to experimental data in order to demonstrate its soundness.

***Keywords*-Integrated circuits, fault attacks, EM fault injection, fault model**

I. Introduction

Electromagnetic Fault Injection (EMFI) has first been suggested as a threat against secure circuits in [1], [2]. Despite this early warning, first concrete results were published [3] in 2007. In this work, authors demonstrated that a low cost EMFI is efficient to apply a Bellcore attack [4].

After these seminal works, [5], [6] introduced an alternative EMFI setup built around a high voltage pulse generator and a coil with a ferrite core [7]. The interest of such a system over the one used in [3] being its high time resolution rendering induced faults more repeatable and controllable. Following this work, several papers used similar systems [8], [9], [10] to evaluate the effectiveness of EMFI for performing various attacks on different ICs or again to analyse the impact of EMFI on embedded codes.

In addition to the proposal of this EMFI system, [6] suggested EM induced faults are timing faults due to the disruption of the power and ground voltages by EM pulses. Following this suggestion, authors of [11], [12], [13] proposed EM pulse detectors which operation are based on the monitoring of propagation delays or other timing metrics.

However, in 2014, [14] experimentally demonstrated that the timing fault model was not enough to explain how EMFI induces faults. To that end they showed EMFI induces faults in a circuit at rest (clock stopped), i.e. in an IC in which timing faults cannot be induced. They finally demonstrated in [15] that EM faults are not timing faults before proposing from their numerous experiments a model called the sampling fault model [16]. This model states that EMFI induces faults by directly disrupting the inputs (D, CK , $reset/set$, Vdd and Gnd) of D-type Flip-Flop (DFF) while they are switching. This implies as it has been experimentally observed and as depicted Fig.1 that EMFI susceptibility varies in time and is greater around the rising edges of the clock. More precisely, [16] claims EM susceptibility is higher during time windows centered on the rising clock edges. In addition, it is also stated that the duration of these time windows is independent of the clock frequency at which operate ICs. If experiments have

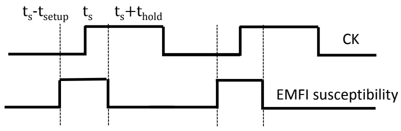

Figure 1. The sampling fault model introduced in [16]

been demonstrated in accordance with this model, up to now and up to the best of our knowledge, this experimental model remains unexplained even if a fully digital EM pulse detector derived from it has been proven experimentally efficient [17].

Within this context, this paper aims at explaining how EM faults occur. This explanation is conducted by successively modeling the interaction between EM probes and ICs, EM pulses on ICs and finally by analyzing in detail the impact of EM pulse on their operation. We do believe that the proposed model could be of great help to design low cost hardware countermeasures as well as to derive design guidelines to mitigate EM induction effects.

This paper is organized as follows. Section II derives from basics about EM induction a modeling of the interaction between EM probes and ICs. This modeling is then used to identify the transient effect of EMFIs on the power and ground voltages of ICs. The transient perturbation induced by EMFIs known, its impact on the operation of ICs is analyzed to identify how EM faults occur in section III. Section IV gives experimental evidences of the proposed model soundness before drawing a conclusion in section V.

II. Impact of EMFI on ICs supply voltage

The idea on which EMFI relies is the generation of a sudden variation of the magnetic field in neighborhood of the target IC surface, such a variation being likely to induce

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parasitic currents disrupting its operation. Therefore, EMFI is a direct exploitation of EM induction principle. Next paragraphs recall what is EM induction and derive some consequences when used to disrupt ICs.

A. EM induction basics and consequences

The theory of EM induction was developed by M. Faraday in 1831. The latter first experimentally demonstrated an electromotive force, ϵ, is induced in any wire loop when the magnetic field crossing the surface it encloses changes. Finally he demonstrated ϵ is proportional to the derivative of the magnetic flux crossing the aforementioned surface. The direction of ϵ is given by Lenz's law which states that an induced current flows in the wire loop in the direction that opposes the change which produced it.

The first key lesson of M. Faraday's observations is that EMFI is expected to induce a parasitic current in all wire loops in ICs, with amplitude proportional to the variation rate of the magnetic field. The effectiveness of EMFI in inducing fault being proven, the important question is what are the wire loops one can find in an IC.

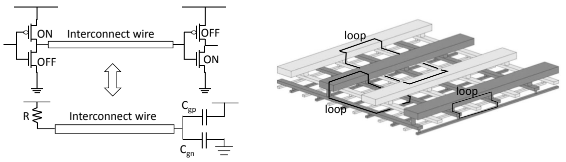

Figure 2. Interconnect wire (left) and 3D power and ground meshes (right)

There are two types of wires in ICs with different roles and characteristics: interconnect wires and power/ground rails. As depicted Fig.2, interconnect wires carries an electrical signal from the output of a CMOS gate to the input(s) of one or several CMOS gates. Therefore, at one of its end an interconnect wire is connected to either a power or ground rail through a transistor under conduction being equivalent to a low resistance. At the other end it is connected to the input(s) of one or several CMOS gates i.e. at MOS capacitances of extremely high (infinite in theory) resistance. Hence, interconnect wires do not form wire loops or extremely resistive wire loops. They are thus not prone to current variation induced by EM induction. Contrarily to interconnect wires, power and ground rails form two 3D meshes delivering the Vdd and Gnd signals to CMOS gates. These two meshes, Fig.2, encompass many vertical and horizontal wire loops and are thus extremely prone to EM induction.

From these simple considerations one can assume with a high confidence level that EM disturbances mainly alter the behavior of the power and ground networks. The resulting question is then how to model the interaction between an EM probe and both the power and ground meshes. This question is addressed in the next subsection.

B. EM coupling modeling

From the above considerations related to wires, the interaction between coils used to perform EMFIs and ICs reduces not to one EM coupling but to two : one between the coil and the Vdd network and the other between the coil and the Gnd network. This interaction can be modeled [18], [19] as depicted Fig.3a.

Fig.3a models the EMFI probe with its inductance L_{probe}, its resistance R_{probe} and its capacitance C_{probe}. This probe is connected to a generator delivering a pulse of amplitude V_{pulse}, of width PW with rising and falling edges of duration T_r and T_f. The two EM couplings are modeled using two mutual inductances M_G and M_V sharing L_{probe}. The values of these mutual inductances [20] are given by:

$$\begin{aligned} M_G &= k_G \cdot \sqrt{L_{probe} \cdot L_G} \\ M_V &= k_V \cdot \sqrt{L_{probe} \cdot L_V} \end{aligned} \tag{1}$$

where L_G and L_V are inductances associated to the part the power and ground networks facing the EM probe.

One could expect to have $L_G \simeq L_V$ because of nearly symmetric routings of the power and ground networks in ICs. However, even if $L_G \simeq L_V$ the mutual inductances M_G and M_V are not expected to be equal because of the probe positioning above the target IC surface but also because Vdd and Gnd rails can be routed on different metal layers. This asymmetry in the EM couplings is modeled by k_G and k_V with values in $[0, 1]$.

With such a modeling, an EMFI induces two electromotrice forces (emf), one in the Vdd mesh and one in the Gnd mesh which effects are modeled by two potential differences between V_1 and V_2 and G_1 and G_2. At that stage, the next point is to study the effect of these emf.

C. Impact of EMFI on the power and ground bias

To understand EMFIs effect on power and ground networks, noise free electrical simulations were performed with SPICE. Prior to these simulations, the whole situation has been modeled. The resulting model is depicted Fig.3. It incorporates the pulse generator, the power source (VDC), the EMFI probe, the EM couplings. It also incorporates the power and ground networks with their

- the power and ground pads modeled by resistances R_{pad}, inductances L_{Bond} (that of bondings) and capacitors C_{decap} (the decoupling capacitors).
- the power and ground grids described using a distributed RC model with C_{GV} the coupling capacitors between the Gnd and Vdd rails and R the wire resistance.

After this modeling step, simulations were launched. Fig.3c depicts what has been observed in the specific case of an EMFI with $V_{pulse} = 400V$, $PW = 10ns$. This specific case, reported as an example of what has been observed during many simulations done with various parameters, gives the

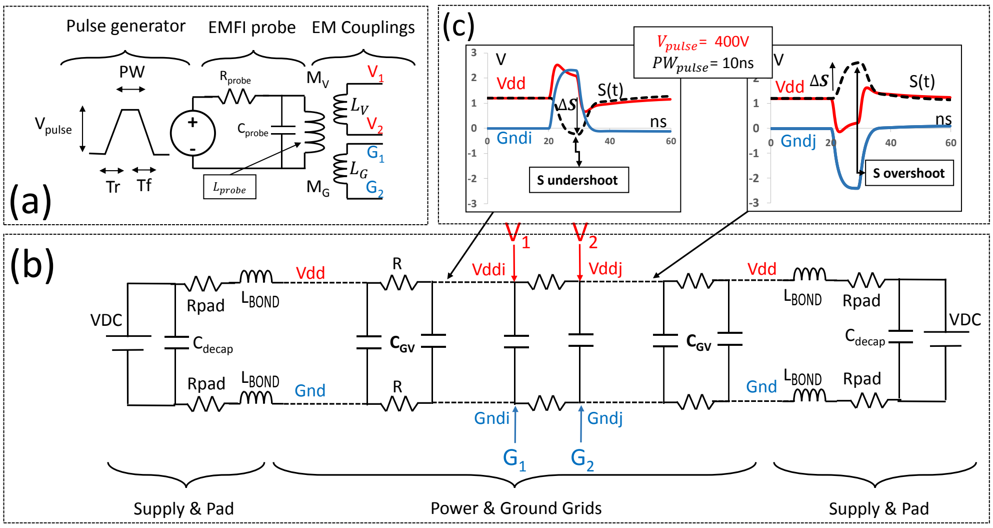

Figure 3. Complete model for analyzing the impact of EMFIs on the power and ground networks. (a) EM probe and coupling models, (b) model of the power / ground networks (c) effect of EMFIs on Vdd and Gnd voltages. (Typical parameters values : $R = 12\Omega$, $C_{GV} = 100fF$, $L_{BOND} = 5nH$, $R_{pad} = 10m\Omega$, $C_{Decap} = 10nF$, $L_{probe} = 10nH$, $L_V = L_G = 15pH$, $R_{probe} = 0.2\Omega$, $C_{probe} = 1nF$)

typical waveforms of $Vdd(t)$ and $Gnd(t)$ along the power and ground grids red and blue, respectively. It also gives in black the typical waveform of the voltage swing:

$$S(t) = Vdd(t) - Gnd(t) \tag{2}$$

Fig.3c clearly shows that the considered EMFI induces voltage drops on both Vdd and Gnd rails on the right side of the power / ground grids (right of $V2$ and $G2$) while voltage bounces are induced on the left of $V1$ and $G1$. The amplitude ΔS of these perturbations increases with V_{pulse}. This is a direct effect of the emf that call electrical charges from the right part of the power / ground grid and push them to the left.

Fig.3c also shows that, because of the asymmetrical coupling ($k_G = 3 \cdot k_V$), the drops/bounces on Vdd and Gnd have different amplitudes. This results in undershoots and overshoots of voltage swing S that propagate toward the pads while being attenuated along their travel. For symmetrical EM couplings the voltage swing ($\Delta S = 0$) remains stable at its nominal value because drops/bounces on the two rails have exactly the same shape. In this unlikely case no fault can occur.

III. Impact of EMFI on ICs operation

At that stage, one may wonder about the conditions EMFIs must met to induce sampling faults and how sampling faults occur. These questions are addressed in the next sections.

A. Simulated Circuit

Following the empirical sampling fault model, we simulated the effects of these swing undershoots/overshoots on the circuit depicted Fig.4. It represents the end of a logic path arriving at the input of a DFF. It is thus a common structure that can be found thousand of times in ICs. In this figure, 'Glue logic' blocks are chains of 20 inverters while the 'clock tree' block is made with 4 clock inverters. C_D and C_{CK} are additional loads modeling the inputs of additional gates connected at this point but also interconnect loads.

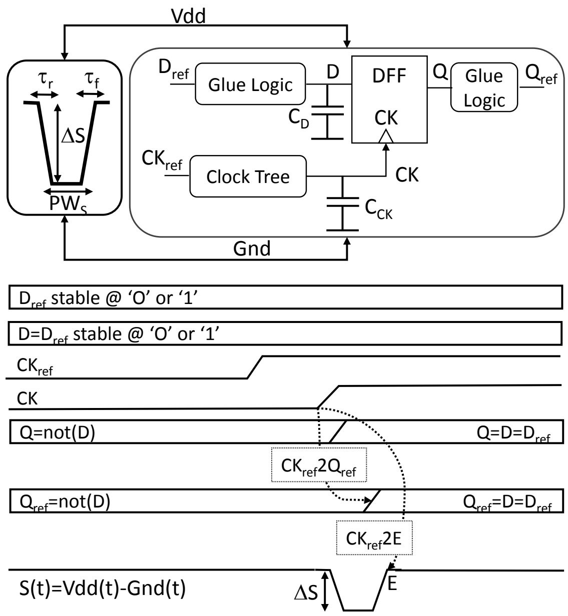

Figure 4. Top : circuit considered during simulations, Bottom: waveforms of signals during EMFI effect analysis

Because this circuit is small, its elements are expected to be placed and routed in a reduced area compared to EM probe size. We therefore assumed that all its elements simultaneously experience the same swing undershoots or overshoots.

Still following the sampling fault model stating that EM induced faults are not timing faults we did consider the signals, D_{ref} and D stable at the same value '0' or '1' and we did observe only what occurs before and after a single rising clock edge. Therefore if faults occur in simulation they are not timing faults but sampling faults.

B. IC operation criterion

Because we had planned to explore the effect of many parameters related to either to the EMFI settings or to the circuit itself, a figure of merit to estimate the effect of EMFIs on the circuits has been set up.

In absence of any perturbation, as illustrated Fig.4, the DFF copies at the rising edge of CK the data D onto Q and then Q propagates until Q_{ref}. The times spent by the data to propagate up to Q and Q_{ref} in normal operating conditions are denoted $CK2Q|_{nom}$ and $CK2Q_{ref}|_{nom}$. Considering that EMFIs alters this operation we did consider the figures of merit F and F' defined as :

$$F = \frac{CK2Q|_{nom}}{CK2Q|_{EMFI}} \quad F' = \frac{CK2Q_{ref}|_{nom}}{CK2Q_{ref}|_{EMFI}} \tag{3}$$

with $CK2Q|_{EMFI}$ and $CK2Q_{ref}|_{EMFI}$ the propagation delays from CK to Q and Q_{ref} in presence of a perturbation. With these definitions, several effects of EMFIs can be differentiated :

- $F(F') > 1$: EMFI accelerates the propagation of the data from D to Q (Q_{ref}),
- $F(F') = 1$: EMFI has no effect on IC operation,
- $1 > F(F') > 0$: EMFI slows down the propagation of the data from D to Q (Q_{ref}) and can induce timing faults at the next rising clock edge,
- $F(F') = 0$: D is not propagated from D to Q (Q_{ref}). This corresponds to the occurrence of a sampling fault.

C. Effect of EMFI

To observe the effect of EMFIs on the ICs operation, we launched many simulations with different values of ΔS, PW_S, τ_r and τ_f (see Fig.4) and $CK_{nom}2E$; $CK_{nom}2E$ being the time separating the nominal (without perturbation) arrival time of CK rising edge (denoted CK_{nom}) from the ending (denoted E) of the swing perturbation as depicted Fig.4. For sake of simplicity this paper gives only results related swing undershoots and $D_{ref} =' 1'$. Similar results can easily be obtained for swing overshoots and D_{ref} stable at the value '1'.

Fig.5 gives, for ΔS ranging between $0V$ and $2.2V$, the evolution of F wrt $CK_{nom}2E$. During these simulations, the

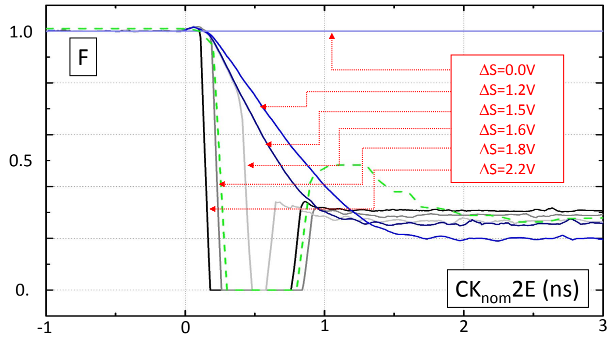

Figure 5. F vs $CK_{nom}2E$ for $\Delta S \in [0V, 2.2V]$, $PW_S = 5ns$, $\tau_r = \tau_f = 2.5ns$. The continous curves are for $D = 1$ and a swing undershoot. The dashed green curve is for $D = 0$ and a swing overshoot.

pulse width, PW_S, was set to $5ns$ and the transition times, τ_r and τ_f, of the swing undershoot were set to $2.5ns$.

For ΔS lower than 1.5, F is strictly positive. No sampling fault occurs. For larger ΔS values $F = 0$ for a while. Therefore there is a threshold for V_{pulse} (and therefore ΔS) below which no sampling fault is induced.

For a powerful EMFI ($\Delta S = 2.2V$), F has values between '0' and '1' for $CK_{nom}2E \in [-1; 0.2] \cup [0.7; 3]$ns. Therefore, for such settings, EMFI induces an increase of the delays $CK2Q$ and $CK2Q_{ref}$. Regarding $CK2Q$, the increase consists in roughly a triply of the DFF propagation delay ($\simeq 150ps$). Concerning $CK2Q_{ref}$, the delay is multiplied by 1.7 (less for lower ΔS) as illustrated Fig.6.

According to the operating frequency of the DUT and to the timing slack introduced at the design step to take Process, Voltage and Temperature variations into account, these increases can induce timing faults at the next rising edge of the clock. However this is unlikely to occur when characterizing a chip under its nominal voltage because timing margins (especially those considered for voltage variations) are greater (especially for ICs operating with clock frequency like micro-controllers) than the timing margins [21] for advanced technologies. This could explained why timing faults were not reported in [16] and why we did not observe timing faults during the experiments reported in section IV. However the situation could change if one lower the supply voltage during EMFI characterizations.

Fig.5 also indicates that between $CK_{nom}2E = 0.2$ns and 0.7ns, F suddenly falls down to 0. There are thus, as observed in former works, time windows during which sampling faults occur. These time windows are denoted by Sampling Fault Window (SFW) afterward. Surprisingly, sampling faults occur during the rising (second) edge of the swing and not during the falling (first) edge. This means sampling faults occur during the recovery phase of the supply voltage.

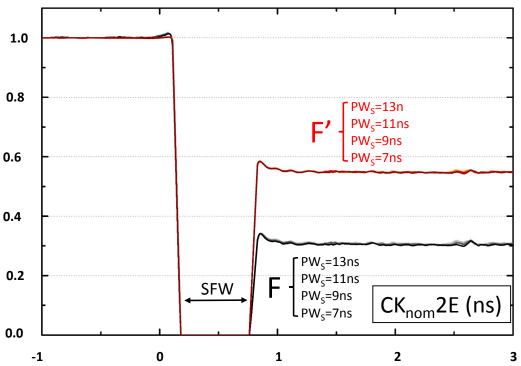

Figure 6. F and F' vs $CK2E$ for $\Delta S = 2.2V$, $PW_S = 7, 9, 11, 13ns$ and $\tau_r = \tau_f = 2.5ns$

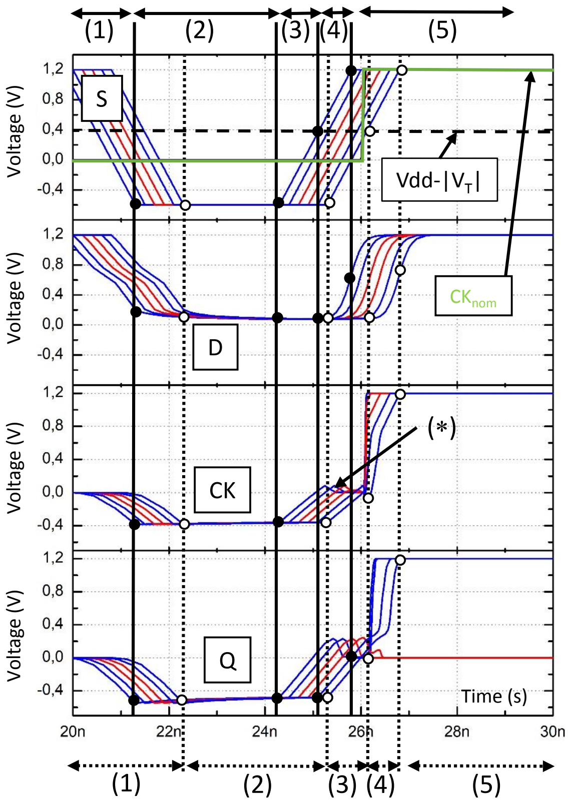

Figure 7. Waveforms of S, D, CK and Q for several values of $CK_{nom}2E$ with $PW_S = 3ns$ and $\tau_r = \tau_f = 1.5ns$

D. How sampling fault occur

The modeling approach has shown EMFIs can induce sampling faults. This is in accordance with what has been observed in practice in former works and sustains our simulation flow methodology.

At that stage, the next key point is to understand how they occur. To that end, Fig.7 reveals the waveforms of signals S, D, CK and Q for several values of $CK_{nom}2E$. In this figure, red (blue) waveforms correspond to a case for which sampling faults occur (resp do not occur).

As depicted, the EMFI effect can be divided into 5 regions which positions wrt CK_{nom} (in green) varies with $CK_{nom}2E$. Region (1) corresponds to the first edge of the voltage pulse delivered to the EM probe and therefore to the first EM pulse. In this region the supply voltage is quickly reversed by induction currents. As a result, all signals falls down to zero or even lower because of capacitive couplings between the related nets and Vdd and Gnd.

At the end of this first edge of V_{pulse}, the IC stalls and remains quasi stable all along region (2). Region (3) starts with the rising edge of the voltage pulse (producing a second EM pulse of opposite polarity) and ends when the $S(t)$ reaches $Vdd - |V_T|$. This phase corresponds to the beginning of the supply voltage recovery. However, because $S(t)$ is lower than the threshold voltages $|V_T|$ of transistors (assumed equal for N an P transistors for sake of simplicity), all signals remain stable.

Region (4) starts when $S(t)$ crosses $Vdd - |V_T|$. Transistors are turned on and the structure carries on its recovery and converges toward a final state with the occurrence or not of a sampling fault according to the $CK_{nom}2E$ value. Finally, when the supply voltage reaches back its nominal value the structure enters region (5) and operates nominally.

One key observation is that sampling faults occur during the recovery phase of the supply voltage. This highlights the importance of using voltage pulse generator with fast rising and falling edges so that to induce an EM pulse with two opposite polarities. This ensures EMFI with high time resolution. The role of the first pulse is to quickly reverse the supply voltage while that of the second is to quickly restore it.

To understand why in some cases a sampling fault occurs, we have to look at the internal operation of the DFF. To that end, Fig.8 gives a zoom on what is going on in the DFF in regions (3) to (5). It reports the nominal waveform of CK (in green), and disrupted waveforms of S, D, Q and those of two DFF internal signals.

The first internal signal is CK_I. It is the internal clock signal of the DFF, the one triggering its master and slave. In practice, it is just a delayed copy of CK by an internal buffer which role is to ensure fast internal clock edges and therefore reduced setup and hold times. The second internal signal is the value stored in the master of the DFF, i.e. the output of the first DFF stage which is a tristate gate. In normal operation, the stored value is equal to $not(D)$ ('0' in our case) after the rising edge of the CK_I.

Fig.8 shows that three cases (black, red, blue) must be distinguished according to $CK_{nom}2E$. Sampling faults occur only for the red case.

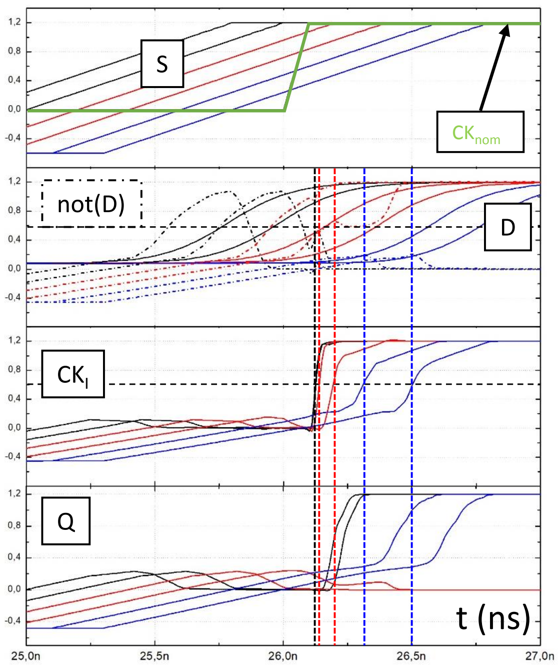

Figure 8. Waveforms of S, D, Q, CK_I and $not(D)$ for several values of $CK_{nom}2E$ with $PW_S = 3ns$ and $\tau_r = \tau_f = 1.5ns$

In the first case (in black), the recovery of $S(t)$ ends before the nominal arrival time of CK in green. In that case, D has time to settle back to a value higher than $0.5 \cdot Vdd$ before the arrival of the internal clock edge CK_I. As result, $not(D)$ is lower than $0.5 \cdot Vdd$ and the DFF samples the right value. In the second case (in red), the recovery of the $S(t)$ is well started but not achieved when the normal clock edge should have occur. In that case, D has little time to recover its expected value. As result, $D < 0.5 \cdot Vdd$ and $not(D) > 0.5 \cdot Vdd$ when CK_I rises as shown Fig.8. Consequently the DFF samples the wrong value. The third case (in blue) is quite surprising. It corresponds to a recovery phase starting just before or after the nominal arrival time of CK. In that case, D and $not(D)$ starts rising all together following $S(t)$. However, because CK_I recovery is faster than that of D and $not(D)$, the DFF stores a '0' in the master i.e. stores the right value. However, the way this correct value has been sampled by the DFF is completely abnormal. It is the result of the simultaneous wake up of all signals and not of a correct operation of the DFF.

These three cases explain why and how EMFI induces sampling faults only during short time windows and sustains the experimental observations reported in [16]. They also explain why timing settings of EMFI are so important in practice.

The concordance between simulations and experimental results given in [16] sustains the soundness of the EM fault model described in this paper. However, to further demonstrate it, comparisons between experimental and simulation results are given in the next section.

IV. Experimental validation

Getting direct experimental evidences demonstrating the validity of the proposed model is extremely difficult because EMFIs pollute all measurements at several meters. One alternative is to integrate embedded sensors in DUT. However, even this complex approach is insufficient because the supply voltage of these embedded sensors will be disrupted by EMFIs.

We therefore searched for indirect evidences of the soundness of the EMFI model and simulation flow methodology introduced in this paper. The idea has been to compare trends observed in practice when changing EMFI settings with trends reported by simulations when varying the same parameters. Among the parameters that can be easily varied in practice one can find :

- the clock frequency, F_{CK}, of the device under test,
- the amplitude, V_{pulse}, of the voltage pulse
- the width, PW, of the voltage pulse.

Next sections give experimental and simulation data related to variations of those parameters but also provide detail about the DUT and the EMFI platform we used.

A. EMFI platform

Our EMFI platform features a voltage pulse generator delivering pulse of amplitude ranging between $\pm 50V$ and ± 400. The pulse width of the generated pulse can be as low as $6ns$ and as high as $100ns$. The timing accuracy of the platform is equal to $\pm 100ps$ + 0.03% of the pulse generator latency. The latter, measured between the external trigger and the shot, can be tuned by step of $10ps$ between $370ns$ and $1s$. Used EM probes are coils with spaced turns around a ferrite core with a diameter ranging $300\mu m$ and $1500\mu m$. Fig.9 shows an EM probe above our testchip.

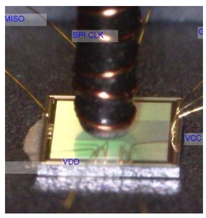

Figure 9. EM probe above the frontside of our testchip

B. Device Under Test and EMFI

EMFIs were applied to our DUT, designed with a low power $40nm$ CMOS technology. The core of this testchip occupies an area equal to $2150\mu m \times 2150mm$ and is supplied with $1.2V$. The clock frequency of this testchip can be externally controlled.

This testchip embeds among various crypto-primitives an hardware AES which was the target of our EMFI campaigns. These EMFIs were performed with various settings so that to verify if sampling fault windows exist and behave as predicted by the modeling with respect to F_{CK}, V_{pulse} and PW.

C. *Effect of F_{CK} variations*

To verify the existence of SFW, EMFIs were performed at different EM probe positions and different values of F_{CK}. These EMFIs were done so that EM shots sweep at least two clock periods with a time step equal to $100ps$. For each considered shot time, t_{shot}, 100 EMFIs were performed so that to estimate the probability P_f to induce a fault. Fig.10 gives the evolution of P_f vs t_{shot} for three values of F_{CK}.

As expected SFW appear on the probability (P_f) traces. These windows repeat, accordingly to the model, with a period equal to that of the DUT ($30MHz$, $50MHz$ and $70MHz$). Their width measured at $P_f = 0.5$ are as expected independent of the clock frequency. However their value ranges between $6.3ns$ and $7.1ns$. This uncertitude in the measure of the SFW width is mainly due to the pulse generator jitter ($\pm 100ps + 0.03\%$ of the tuned latency) which has a value close to $0.9ns$ for the set latencies allowing to shoot in the AES at the considered clock frequencies. Therefore, there is a quite good agreement between the model prediction and experiments.

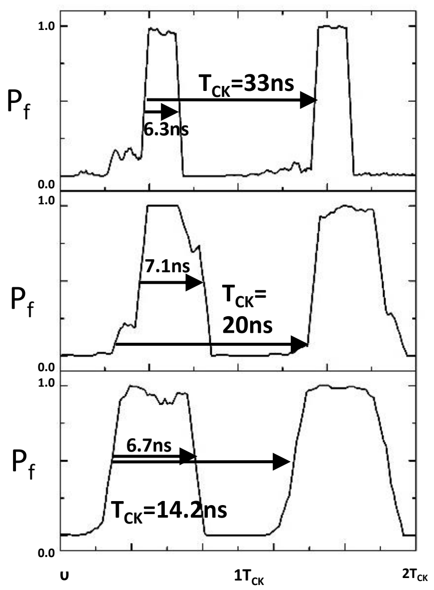

Figure 10. Probability of inducing a fault vs normalized time ($\frac{t}{T_{CK}}$)

D. *Effect of V_{pulse} variations*

Other similar experiments and simulations were done to estimate the evolution of SFW width wrt V_{pulse}. Simulations predict the width of sample fault windows increases with V_{pulse}. This trend can also be observed experimentally as reported Fig.11. This figure gives the normalized simulated and measured trends of SFW width with V_{pulse}. Normalization has been done with respect to the SFW width measured and simulated for $V_{pulse} = 380V$. The agreement between simulations and experiments demonstrates the soundness of the proposed EM fault modeling.

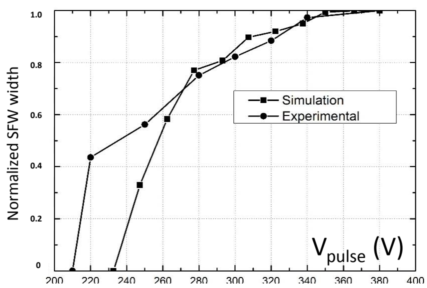

Figure 11. Simulated and measured normalized SFW width wrt V_{pulse}

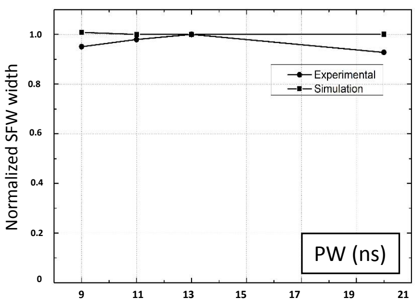

Figure 12. Simulated and measured normalized SFW width wrt the voltage pulse width PW

E. *Effect of PW variations*

Finally, similar experiments than those described in previous section were done to analyze the effect of PW on the SFW width. From simulations (see Fig.6) it was expected that PW does not affect much the SFW width. Experiments have confirmed this observation. To further sustain this result, Fig.12 gives the normalized simulated and measured trends of SFW width wrt PW. Normalization has been done wrt the value obtained for $PW = 13ns$ Again, the agreement between simulations and experiments is good. This demonstrates the soundness of the proposed EM fault modeling.

V. CONCLUSION

An explanation of how EMFI could induce faults has been given in this paper. The associated simulation approach has been used to forecast how EMFI settings changes the probability to induce faults. The simulated trends have been successfully confronted to experiments. This model states that EM faults are induced because voltage pulse applied to EM probe generates an EM pulses with two opposite polarities. The first one induces a reversal of the supply voltage while the second one restores it. During the supply voltage recovery, the DFF can sample wrong values because they operates under abnormal conditions.

The accordance between the simulation and experiments shows that the model is headed in the right direction. However, the model covers only cases in which the EMFI are shorter than the clock period (case of microcontrolers) and applied to glue logic. Furthermore, EMFI effects can vary significantly between devices or EM configuration, so this explanation given in this paper may not be the only or final one. There is thus room for refinements. Further works will focus on refining the proposed model (taking the effect of the substrate into account, considering EMFI effect on analogue blocks, etc) but also on extending it to cases for which the EM induced perturbation lasts for more than a clock cycle.

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